

Aswanth Manoharan

8 Costguard Terrace, Bray, Ireland | aswanthanu2024@gmail.com | +353 877617142 | LinkedIn | GitHub | Portfolio

Summary

Software Engineer with experience in data validation, analysis, and backend systems in cloud-based environments. Strong proficiency in Python, SQL, and data processing, with hands-on experience in ETL pipelines and data quality assurance. Proficient in AWS services including Lambda, S3, EC2, and IAM, along with CI/CD workflows and containerized deployments using Docker.

Detail-oriented and process-driven, with a strong focus on data accuracy, system reliability, and documentation in collaborative environments. Experienced with scalable and serverless backend architectures in production systems, applying Agile methodologies.

Experience

Python Backend Developer (Part-Time) - HitoAI, Dublin, Ireland Jun 2024 – May 2025

- Developed and maintained 4 FastAPI services supporting internal tools, ensuring reliable and consistent data flow across backend systems using Python.
- Monitored application logs and validated API data outputs, identifying inconsistencies and reducing intermittent failures by 15%.
- Deployed containerized applications on AWS (EC2, S3) using Docker, supporting scalable backend services handling 2,000+ daily API requests, demonstrating cloud and DevOps skills.
- Configured AWS S3 storage and IAM roles to enforce secure and structured data access control.
- Contributed to CI/CD pipelines using GitHub Actions, improving deployment reliability and reducing manual effort by approximately 30%.
- Worked with cloud-based backend workflows demonstrating exposure to scalable system architectures.
- Collaborated with frontend components built in React, ensuring smooth API integration and data flow between backend services and UI layers.

Junior Python Developer - Tfora Technologies, Kerala, India Mar 2023 – Apr 2024

- Built and maintained REST APIs using Django and Django REST Framework, ensuring accurate and structured data exchange between systems with Python.
- Optimised SQL queries and performed data validation checks, improving API response times by 20%, demonstrating strong SQL skills.
- Worked with structured datasets to support feature development, ensuring consistency and correctness of application data.
- Implemented structured Git/GitHub workflows including pull requests and code reviews.
- Delivered features on bi-weekly release cycles within an Agile/Scrum environment, applying DevOps practices.

Machine Learning Intern - Ignitus, Remote Sep 2022 – Feb 2023

- Processed and analysed 50,000+ record datasets, improving data preparation efficiency by 15%.
- Applied machine learning algorithms including SVM and logistic regression, improving accuracy by 18%.
- Built reproducible ML pipelines using Scikit-learn and PyTorch.
- Performed feature engineering and preprocessing for structured and text data.

Projects

Real-Time Fall Detection System Using YOLOv11 2025

Tech Stack: Python, PyTorch, Ultralytics YOLOv11, OpenCV, Django, WebSockets

- Fine-tuned a YOLOv11 object detection model, ensuring high accuracy with 98.9% recall.
- Integrated RTSP live camera streaming into a Django web application, enabling real-time data processing.
- Designed modular training and evaluation pipelines for reproducible ML workflows.

ETL-Driven Data Warehouse Design and BI Reporting for Retail Analytics 2024

Tech Stack: SQL Server, SSIS, Tableau, Excel

- Designed a star schema data warehouse supporting structured analytics.
- Built automated ETL pipelines processing 65,000+ records ensuring data consistency.
- Developed interactive dashboards using Tableau for business insights and reporting.

E-Commerce Platform (Django + React) 2024

Deployed on Azure VM

- Built a full-stack e-commerce platform using Django (backend) and React (frontend), implementing authentication, search, filters, and order management.
- Designed backend APIs and database schema for scalable data handling.
- Deployed application on cloud infrastructure using AWS EC2, demonstrating production deployment experience.

Skills

Programming: Python, SQL, Java (Basics), JavaScript (Basics)

Cloud & Serverless: AWS (Lambda, S3, EC2, IAM), Docker, CI/CD (GitHub Actions)

Frontend: React, JavaScript, HTML, CSS, Bootstrap

Backend & APIs: FastAPI, Django, Django REST Framework

Data & ML: Pandas, NumPy, Scikit-learn, PyTorch, NLP

Data Engineering: ETL Pipelines, Data Warehousing, SSIS

Monitoring & DevOps: Prometheus, Grafana, Git, GitHub Actions

Core Strengths: Problem Solving, System Design, Debugging, Data Processing, Communication, Fast Learning

Education

MSc in Data Analytics - Dublin Business School

2024–2025

BSc in Chemistry & Industrial Chemistry - University of Kerala

2017–2020